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Device for pneumatic launching of a drone

Abstract

A device for pneumatic launching of a drone includes a launch tube, a pressurization chamber situated at the rear of the tube and provided with a passage opening leading into the tube, and a device for temporarily closing the passage opening. The temporary closing device is arranged to change from a closed configuration to an open configuration instantaneously under the action of the gas pressure inside said chamber, after the latter has reached a trigger pressure, and the cross section of the passage opening is sufficiently large that, instantaneously with the change to the open configuration, the gas pressure prevailing in the chamber and the space located behind the drone, then in communication with each other, is the desired launching pressure for the drone.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3040628	12/1961	Sieve et al.	N/A	N/A
3968947	12/1975	Schlegel et al.	N/A	N/A
7584925	12/2008	Miller et al.	N/A	N/A
7739938	12/2009	Nair	89/7	F41B 11/62
8505430	12/2012	Miralles	89/1.816	F41F 3/042
9453705	12/2015	Sylvia	N/A	F42B 15/105
9470477	12/2015	Su	N/A	B64F 1/04
9612085	12/2016	Powell	N/A	F41F 3/07
10464693	12/2018	Sylvia	N/A	B63G 8/28
11591050	12/2022	Graham	N/A	B64U 80/70
11959725	12/2023	Faizan	N/A	F41F 3/07
12030666	12/2023	Kuperman	N/A	B64F 1/04
12111137	12/2023	Theiss	N/A	B64U 70/30
2012/0205488	12/2011	Powell	244/63	B63G 8/30
2014/0150683	12/2013	Dunaway	102/530	F41F 3/04
2018/0244402	12/2017	Kahlon	N/A	B64F 1/04
2023/0049500	12/2022	Murphy	N/A	B64U 70/50

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
116176898	12/2022	CN	B64F 1/06
118145048	12/2023	CN	B64U 70/50
2 297 770	12/1975	FR	N/A
3133051	12/2022	FR	B64F 1/06

2023161099	12/2022	KR	B64C 39/024
2013/011412	12/2012	WO	N/A
2014/035518	12/2013	WO	N/A
2015/127178	12/2014	WO	N/A

OTHER PUBLICATIONS

Sep. 20, 2022 Search Report and Written Opinion issued in French Patent Application No. 2201677. cited by applicant

May 24, 2023 International Search Report issued in International Patent Application No. PCT/IB2023/051711. cited by applicant

May 24, 2023 Written Opinion issued in International Patent Application No. PCT/IB2023/051711. cited by applicant

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The technical field of the invention is that of pneumatic launch devices, and more specifically, devices for pneumatic launching of drones.

(2) Drones can carry payloads for civil or military surveillance, intelligence, combat or transport missions. As small flying machines that are cheaper and simpler to operate than aircraft with pilots, drones are enjoying a boom.

Description of Related Art including information disclosed under 37 CFR 1.97 and 1.98.

(3) Several means for launching drones are known, including pneumatic launch devices.

(4) A pneumatic launch device is one that uses compressed air, or any other type of pressurized gas, to cause the drone to be launched.

(5) Typically, a pneumatic launch device comprises a pressurization chamber which is pressurized by a compressed gas. Releasing the pressure in the chamber generates energy, which is used to propel the drone.

(6) There are many types of pneumatic launch devices. In particular, pressure can be released by a variety of means.

(7) A known type of pneumatic launch device is a drone launcher with a pneumatic catapult, such as that disclosed, for example, in French patent application FR2297770 A1. The operating principle of this pneumatic catapult is the expansion of a pressurized gas in a telescopic assembly of two concentric cylindrical tubes so as to cause one to move rectilinearly relative to the other.

(8) Another known type of pneumatic launch device is a device with a launch tube, such as those disclosed in US patent U.S. Pat. No. 7,584,925 B2 and in international patent applications WO2015/127178, WO2013/011412 and WO2014/035518.

(9) The aim of these devices with launch tube is to propel the drone a few metres above the vehicle carrying the launch tube, for deployment of the drone, in order to avoid any contact between the drone and the antennae mounted on the vehicle and to deploy the drone quickly.

(10) In the launch device according to U.S. Pat. No. 7,584,925 B2, during launch, a launch gas contained in a launch gas reservoir is released into a launch tube after opening a valve system, a first part of this launch gas forces a free sliding piston mounted inside the launch tube to move

towards the open front end of the launch tube, and a second part of the launch gas leaves the launch tube through its open front end.

(11) In the launch device according to patent application WO2015/127178, during launch, a compressed gas contained in a chamber is released into a launch tube containing the drone after opening a bleed valve.

(12) In the launch device according to patent application WO2014/035518, during launch, a gas generated by a gas generator arranged inside a launch tube is introduced into a high-pressure volume defined between a skirt of a sabot and a casing containing the gas generator, the skirt being releasably coupled to the casing via an adhesive coupling. The resulting gas pressure presses against the skirt until the adhesive coupling breaks, once a certain rupture pressure has been reached in the high-pressure volume. The gas that continues to be generated then pushes the released skirt against the wall of the launch tube to drive the sabot along the launch tube until the drone is ejected from the launch tube. The disadvantage of the devices with launch tube described above is that they do not allow sufficient control of the value of the gas pressure by which the drone is ejected, and therefore do not allow control of the speed and height of the drone ejection, so that it is necessary to provide a safety margin before the drone is deployed in order to prevent it from coming into contact with, for example, the vehicle antennae.

BRIEF SUMMARY OF THE INVENTION

(13) The aim of the present invention is therefore to provide a pneumatic launch device that enables the launching pressure by which the drone is ejected to be controlled, thus enabling a drone to be ejected reliably and reproducibly, and by means of which an effectively quick deployment of the drone can be implemented.

(14) The solution according to the present invention is based on the use of means enabling the pressure by which the drone is to be ejected to be applied instantaneously to the drone, and not a gradual, or even partial, release of gas pressure behind the drone as in devices in the prior art.

(15) Patent application WO2013/011412 describes a device according to the preamble of claim 1.

(16) The present invention thus relates to a device for pneumatic launching of a drone, the device comprising: a longitudinal launch tube having a first, rear end, which is closed and a second, front end, which is open, and the interior of which defines a launch chamber intended to receive a drone and in which there is a seat against which the drone is positioned before it is launched; a pressurization chamber located at the rear end of the launch tube and capable of being pressurized by a pressure source, the pressurization chamber having a first, front end in which is provided a passage opening leading into the launch chamber and, in use, behind the drone placed against the seat; and means for temporarily closing the passage opening, capable of switching instantaneously from a closed configuration, in which the passage opening is closed in a sealed manner, to an open configuration in which the passage opening is fully open,

the temporary closing means being arranged to switch from the closed configuration to the open configuration instantaneously under the action of the gas pressure inside the pressurization chamber, after the latter has reached a predetermined pressure, referred to as the trigger pressure, and the cross section of the passage opening being sufficiently large that, instantaneously with the switching of the temporary closing means from the closed configuration to the open configuration, the gas pressure prevailing in the pressurization chamber and the space behind the drone, then in communication with each other, is the desired launching pressure for the drone, characterized in that the device comprises said pressure source, which is formed by an internal gas source arranged inside the pressurization chamber, said pressure source comprising, placed in the pressurization chamber, a diffuser, delimiting a reservoir with a first reactant, and a second reactant, the first and second reactants, when brought into contact with each other, reacting to generate a gas for pressurizing the pressurization chamber, a solenoid valve being arranged so as to allow selectively a fluid communication between the reservoir and the pressurization chamber.

(17) The expression “fully open” means that the temporary closing means are clear of the passage

opening.

(18) Advantageously, the cross section of the passage opening is between 30% and 50% of the cross section of the launch chamber.

(19) It is possible to provide that the temporary closing means are arranged to switch from the closed configuration to the open configuration by actuating a mechanically actuatable release means. For example, the temporary closing means may thus comprise a flap mounted so as to pivot freely at the outlet of the passage opening, or two flaps mounted so as to pivot freely about axes arranged on either side of the passage opening, and held in a closed position by a mechanical latch movable between a position in which the flap is locked in the closed position, and a release position in which it is disengaged from the flap, which is then caused instantaneously to pivot and open the passage opening by the gas pressure. The movement of the latch can be controlled by any suitable means, such as an electromagnet. In this way, the instruction to launch the drone is implemented by controlling the latch.

(20) In a preferred embodiment, the temporary closing means are arranged to switch from the closed configuration to the open configuration automatically upon the gas pressure inside the pressurization chamber reaching the trigger pressure.

(21) The temporary closing means may then advantageously comprise a rupture disc assembly that is disposed at the passage opening and comprises a calibrated diaphragm extending across the passage opening so as to close it, the diaphragm being dimensioned to break at the trigger pressure.

(22) The calibrated diaphragm may, for example, be made of stainless steel, plastic or graphite. It can, of course, be made of any other suitable material.

(23) The calibrated diaphragm can be designed to break all around its periphery, where it is made integral with the pressurization chamber, or to break from its centre, or in any other rupture mode that allows the passage opening to be fully open.

(24) The rupture disc assembly can be removably fixed to the pressurization chamber. Preferably, the rupture disc assembly comprises a support ring and a fixing ring of the same dimensions and fixed to each other by fixing members passing through holes provided for this purpose in the rings, with the periphery of the calibrated diaphragm sandwiched between the rings, the rupture disc assembly being removably fixed to the front end of the pressurization chamber also by said fixing members.

(25) Such a rupture disc assembly has the advantage of being both low cost and quick to open. When maintaining and reconditioning a new drone, it is simply a matter of replacing the broken diaphragm with a new one, which is more economical than replacing, for example, a pyrotechnic initiator and a gas-generating charge.

(26) The rear end of the launch tube can be formed by a base removably fixed to an inner shoulder of the launch tube, from which base a tubular part extends, the longitudinal axis of which is coaxial with the longitudinal axis of the launch tube and which defines, together with the base, the pressurization chamber, a gasket being interposed between the tubular part and the inner annular edge of the inner shoulder. The device is therefore simple to make and assemble.

(27) The front end of the launch tube can be closed with a tight plug.

(28) Preferably, the first reactant is hydrogen peroxide (H_2O_2) under pressure and the second reactant is a potassium permanganate (KMnO_4)-based solid catalyst.

(29) The device according to the present invention may further comprise a launch sabot capable of being received in the launch chamber and brought into contact against the seat, the launch sabot being configured to contain and surround the drone before launch and during its movement along the launch tube during ejection. The sabot protects and guides the drone in the launch tube during the launch phase, and also guides, pressurizes and distributes the energy developed by the gas pressure.

(30) It should therefore be pointed out here that, according to the present invention, the drone can be in direct contact with the seat of the launch tube, or in indirect contact with it, via the sabot.

(31) Advantageously, the launch sabot is formed of several separable segments, preferably four segments.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

- (1) To better illustrate the subject-matter of the present invention, a particular embodiment will be described below, with reference to the attached drawings. On these drawings:
- (2) FIG. 1 is a side view, in longitudinal cross section, of the pneumatic launch device according to a particular embodiment not forming part of the claimed invention, with a pressure source according to a first embodiment;
- (3) FIG. 2 is a side view, along a first partial longitudinal cross section plane, of the pneumatic launch device of FIG. 1, with a pressure source according to the present invention;
- (4) FIG. 3 is a longitudinal cross section view of the pressure source according to the present invention, along a cross section plane perpendicular to said first cross section plane;
- (5) FIG. 4 is an exploded perspective view of the device of FIG. 1, with the sabot and drone omitted;
- (6) FIG. 5 is an exploded perspective view of the sabot and the drone in their configuration before launch; and
- (7) FIG. 6 is a perspective view of the sabot and the drone in their configuration after launch.

DETAILED DESCRIPTION OF THE INVENTION

- (8) Referring first of all to FIGS. 1, 5 and 6, it can be seen that the pneumatic launch device 1 according to the present invention is intended for launching an object of the drone 2 type.
- (9) Such a drone 2 typically comprises a base body 20 containing a power plant, a battery pack and navigation electronics. The drone 2 shown in FIGS. 5 and 6 comprises two counter-rotating rotors 21a, 21b, i.e. two superimposed propellers. It should be noted that the drone 2 could comprise any number of rotors. The blades 22 of each rotor 21a, 21b are foldable and can therefore be moved from a folded position (FIG. 5) to a deployed position (FIG. 6). The drone 2 is equipped with a payload 23 removably mounted on the base body 20. This payload 23 may be a lethal or non-lethal load. For example, the drone 2 could be equipped with a lethal load of explosive type, a non-lethal load enabling paint or smoke to be released, or an optronic load enabling observation and detection.
- (10) Referring now to FIGS. 1 to 4, it can be seen that the device 1 comprises a launch tube 3 with a longitudinal axis A1, intended to receive the object to be launched, i.e. the drone 2, before it is launched, and ejection means 4 for ejecting the drone 2.
- (11) The launch tube 3 has a generally cylindrical shape about its longitudinal axis A1, and it has a front end 3a forming the mouth for the exit of the drone 2 and a rear end 3b in the region of which the ejection means 4 are arranged.
- (12) It is emphasized here that the terms “front” and “rear” refer to the predetermined direction of movement of the drone 2 relative to the launch tube 3 during launch.
- (13) The front end 3a is closed, prior to launch, by a circular tight plug 30, the diameter of which corresponds to the external diameter of the launch tube 3. An O-ring 31 is interposed between the plug 30 and the internal wall of the launch tube 3, at the front end 3a.
- (14) The rear end 3b is formed by a base 32 in the form of a block of circular cross section, the diameter of which corresponds to the external diameter of the launch tube 3. The base 32 is fixed to an internal shoulder 33 of the launch tube 3, which is annular in shape, by fixing members 34, such as screws, passing through orifices provided in the base 32 and orifices provided in the inner shoulder 33 and located opposite each other.
- (15) The launch tube 3 defines therein a cylindrical launch chamber 35 having, in the region of the rear end 3b, a seat 36, here formed by the inner shoulder 33, against which the drone 2 will be

placed before ejection from the launch tube 3 by the ejection means 4.

(16) The ejection means 4 comprise a pressure source 5 or 5', a pressurization chamber 6 and temporary closing means 7.

(17) The pressure source 5 or 5' is intended to enable the pressurization chamber 6 to be pressurized by compressed gas, which may be air, an inert gas, steam, etc., or any combination thereof.

(18) It is therefore emphasized that, although the term "pneumatic" is used to designate the launch device according to the present invention, it is not limited to the use of air as the gas for launching the drone 2.

(19) The pressure source can be any gas pressure source.

(20) In the embodiment shown in FIG. 1 and not forming part of the claimed invention, for reasons of space, the pressure source 5 is an external pressure source, i.e. it is arranged outside the launch tube 3 and can be brought into communication with the pressurization chamber 6 by means of a pneumatic connection 50. This pneumatic connection 50 has an inlet hole leading into the base 32.

(21) The pressurization chamber 6 is laterally delimited by a tubular part 60, with a longitudinal axis A2, extending perpendicularly from the transverse wall of the base 32, in other words parallel to said longitudinal axis A1, towards the front end 3a of the launch tube 3. The pressurization chamber 6 thus has a rear end 6b closed by said transverse wall and an open front end 6a, and is located inside the rear region of the launch tube 3. The tubular part 60 is here integrally formed with the base 32, but alternatively the pressurization chamber 6 can be fixed to it by any suitable means.

(22) A gas inlet hole 61 is provided in a region of the side wall of the pressurization chamber 6 that is located in the base 32, to allow gas entering through an inlet hole of the base 32 to enter the pressurization chamber 6.

(23) In the embodiment shown in FIGS. 2 and 3, the pressure source is an internal pressure source 5', i.e. located inside the launch tube 3, and more particularly inside the pressurization chamber 6, which is formed here by a pressurization bell 60'.

(24) The pressure source 5' is a chemical initiator, here based on the reaction between hydrogen peroxide (H.sub.2O.sub.2) and potassium permanganate (KMnO.sub.4).

(25) The chemical initiator 5' comprises a diffuser 51 defining a reservoir 52 of H.sub.2O.sub.2 , a solenoid valve 53, a KMnO.sub.4 -based solid catalyst 54, a piston 55, a spring 56 and a reservoir cap 57.

(26) The pressurization bell 60' is in the form of a tubular body intended to receive all the components of the chemical initiator 5' and fixed to the base 32 so as to delimit a chamber similar to that formed by the tubular part 60, namely located at the rear end 3b of the launch tube 3. The longitudinal axis of the bell 60' is thus also coaxial with the longitudinal axis A1 of the launch tube 3.

(27) The diffuser 51 is formed by a generally cylindrical part, one end of which has an annular flange 51a on which the bell 60' is mounted, with the interposition of a gasket between them, and which is made integral with the base 32, in particular by screws, again with the interposition of a gasket between them. The generally cylindrical part comprises, starting from the flange 51a, a rear part 51b in which is provided a cavity leading into the central opening of the flange 51a and receiving the solenoid valve 53, a tubular front part 51c, the front end of which is closed by the reservoir cap 57, and an intermediate part 51d separating the rear and front parts 51b, 51c.

(28) The piston 55 is mounted inside the front part 51c, between the cap 57 and the intermediate part 51d. The spring 56 is a helical spring mounted between the piston 55 and the cap 57, the ends of the spring 56 bearing against the piston 55 and the cap 57, respectively. The cap 57 and the piston 55 each comprise a cavity, said cavities being capable of receiving the spring 56 when the cap 57 and the piston 55 are in contact with each other. The region laterally delimited by the internal side wall of the front part 51c, at the rear by the intermediate part 51d and at the front by

the piston **55**, constitutes the reservoir **52** of H.sub.2O.sub.2 that can be pressurized by the action of the piston **55** and the spring **56**. The front end of the cap **57** is made integral with the front end of the bell **60'**, the latter itself being made integral, in particular by screwing, with the temporary closing means **7**.

(29) The KMnO.sub.4-based solid catalyst **54** is received in the space formed between the first part **51b** of the diffuser **51** and the internal side wall of the bell **60'**. This solid catalyst **54** is therefore in the form of an annular body with a rear wall in contact with the flange **51a**.

(30) As it can be seen more clearly in FIG. 2, first channels **51e** are provided in the intermediate part **51d** so as to place in fluid communication with one another the reservoir **52** and a free space **51f** in the cavity in which the solenoid valve **53** is housed. The first channels **51e** are parallel to the longitudinal axis of the diffuser **51**. Referring now to FIG. 3, it can be seen that second angled channels **51g** are also provided in the intermediate part **51d** so as to place the pressurization chamber **6** in fluid communication with an outlet passage **51h** provided in the centre of the intermediate part **51d** and which can be placed in fluid communication with said free space **51f** by actuating the solenoid valve **53**. The solenoid valve **53** can therefore be controlled so as to cause the contents of the reservoir **52** to enter the pressurization chamber **6** where the solid catalyst **54** is located.

(31) The exothermic chemical reaction between the KMnO.sub.4-based solid catalyst **54** and H.sub.2O.sub.2 contained in the reservoir **52** generates a very large quantity of oxygen and water vapour, allowing the pressurization chamber **6** to be pressurized up to the trigger pressure.

(32) The longitudinal axis A2 of the pressurization chamber **6** is coaxial with the longitudinal axis A1 of the launch chamber **35**. The external diameter of the pressurization chamber **6** is smaller than the internal diameter of the launch tube **3**, so that a free space is formed between the external wall of the pressurization chamber **6** and the internal wall of the launch tube **3**, allowing part of the object to be ejected to be received therein if necessary.

(33) The pressurization chamber **6**, whether defined by the tubular part **60** or the pressurization bell **60'**, has, at its open front end **6a**, an annular internal rim **62** which delimits a circular passage opening **63** leading into the launch chamber **35**.

(34) The temporary closing means **7** are arranged at the passage opening **63** and serve to close it selectively. More specifically, the temporary closing means **7** are able to close the passage opening **63** before launch and leave it free at the time of launch.

(35) In other words, the temporary closing means **7** are able to switch instantaneously from a closed configuration to an open configuration. In the closed configuration, the front end **6a** is closed, so that the pressurization chamber **6** is closed in a sealed manner and no exchange of gas is possible between the pressurization chamber **6** and the launch chamber **35**, which are not in communication. In the open configuration, the passage opening **63** is not closed, the front end **6a** is open, the pressurization chamber **6** and the launch chamber **35** are in communication, and the gas contained in the pressurization chamber **6** enters the launch chamber **35** to eject the drone **2**.

(36) In the preferred embodiments of the present invention (FIGS. 1 to 4), the temporary closing means **7** comprise a rupture disc assembly **70** comprising a calibrated diaphragm **71**.

(37) The rupture disc **70** has a support ring **72** fixed to the front end **6a** of the pressurization chamber **6** and carrying the calibrated diaphragm **71**.

(38) The support ring **72** has a first side, a second side, an outer peripheral edge and an annular inner peripheral edge, the latter delimiting a central opening **72a** the diameter of which, in other words the internal diameter of the support ring **72**, is equal to the diameter of the passage opening **63**. The external diameter of the support ring **72** is equal to the external diameter of the tubular part **60**, and, if applicable, to the external diameter of the bell **60'**. The first side of the support ring **72** is applied against the internal rim **62** and the second side of the support ring **72** is therefore oriented towards the front end **3a** of the launch tube **3**. In other words, the support ring **72** is interposed between the front end **6a** of the pressurization chamber **6** and the launch chamber **35**.

(39) The calibrated diaphragm **71** extends across the central opening **72a** of the support ring **72**. Thus, in the closed configuration, the calibrated diaphragm **71** closes the entire passage opening **63**.

(40) As regards the method of rupture of the diaphragm **71**, various variants may be suitable, provided that the rupture of the diaphragm **71** clears the entire cross section of the passage opening **63**. For example, the diaphragm **71** could rupture at its periphery, i.e. where it joins the support ring **72**, or at its centre.

(41) The rupture disc assembly **70** is fixed to the pressurization chamber **6** by a fixing ring **73**. The fixing ring **73** is a part similar to the support ring **72** and is in the form of a flat annular section of the same dimensions as the support ring **72**. A first side of the fixing ring **73** is applied against the second side of the support ring **72**. Holes, capable of receiving fixing members **74**, are provided in the fixing ring **73**, so as to face corresponding holes provided in the support ring **72** and in the internal rim **62**. The rupture disc assembly **70** is thus fixed to the open end **6a** of the pressurization chamber **6** by fixing members **74** passing through the fixing ring **73**, the diaphragm **71**, the support ring **72** and the internal rim **62**, and the heads of which are located on the launch chamber **35** side. When the rupture disc assembly **70** is assembled, it is therefore interposed between the internal rim **62** and the fixing ring **73**. This type of assembly is easy and enables the rupture disc assembly **70** and the diaphragm **71** to be easily and quickly dismantled after rupture, in order to replace it with another rupture disc carrying an unbroken diaphragm, or to replace the diaphragm alone.

(42) Referring again to FIGS. **1**, **2**, **5** and **6**, it can be seen that the device **1** also includes a launch sabot **8** intended to receive the drone **2**.

(43) The launch sabot **8** is sized and configured to be received in the launch chamber **35** to enable the drone **2** to be guided into the launch tube **3** during the launch phase, and to surround a drone **2** to enable the drone **2** to be protected during launch. It should be noted that a different sabot **8** is defined according to the profile of the drone **2** and its payload **23**.

(44) In the particular embodiment shown, the sabot **8** is formed of several separable segments **80**, in particular four segments **80**. These segments **80** delimit between them a drone receiving space **81** having an opening intended to be located facing the front end **3a** of the launch tube **3**. These segments **80** also delimit between them a receiving space **82** for the pressurization chamber **6**, having an opening intended to allow passage of the pressurization chamber **6** when the sabot **8** is introduced into the launch chamber **35**.

(45) The segments **80** can be identical.

(46) Alternatively, the sabot **8** could be formed of four separable segments, three of which are identical and one of which contains a charging and communication device **9** between the drone **2** and the launch tube **3**. As the sabot **8** is intended to be ejected from the launch tube **3** and to become detached from the drone **2** after launch, the charging and communication device **9** is a wireless device. Preferably, this device **9** is of inductive type and comprises a transmitting coil capable of cooperating with a receiving coil housed in the drone **2**. This device **9** also comprises spring contacts **90** capable of cooperating with a contact plate provided in the launch tube **3** for connection between this device **9** and the launch tube **3**.

(47) The device **1** enables drones to be launched by controlling the launching pressure by which the drone **2** is ejected, and therefore ensures reliable and reproducible ejection for a quick deployment of the drone **2**.

(48) This is possible because: on the one hand, the diaphragm **71** is calibrated to open by breaking when the pressure applied to it by the gas present in the pressurization chamber **6** reaches a predetermined value, which will be the so-called trigger pressure, the rupture disc assembly **70** then switching instantaneously to its open configuration, and on the other hand, the cross section of the passage opening **63** is large enough so that, immediately after the rupture of the diaphragm **71**, the pressure prevailing in the volume composed of the pressurization chamber **6** and the free space of the launch chamber **35** located between the pressurization chamber **6** and the drone **2** (i.e. here the receiving space **82** formed by the sabot **8**), is the desired launching pressure for the drone **2**.

(49) Indeed, at the very moment the calibrated diaphragm **71** breaks, the compressed gas instantly occupies all the free volume available to it.

(50) Consequently, depending on characteristics of the launch tube, such as its internal diameter and length, and of the drone, such as its weight, the speed and the height at which it must be ejected before it can be deployed safely, the launching pressure can be defined in advance, i.e. the gas pressure to be applied to it when it is in position against the seat **36**, taking into account the volume of the pressurization chamber **6** and the volume of said free space, and thus the trigger pressure, i.e. the gas pressure in the pressurization chamber **6** which will allow to obtain the launching pressure after rupture of the calibrated diaphragm **71**.

(51) Thus, when a drone **2** is to be launched, the launch sabot **8**, in the receiving space **81** of which the drone **2** is received, is first inserted through the open front end **3a** of the launch tube **3**, until it is placed in contact with the seat **36**, then the front end **3a** of the launch tube **3** is closed by the tight plug **30**.

(52) When it is desired to launch the drone **2**, the pressure source **5** or **5'** is brought into communication with the pressurization chamber **6** or actuated to pressurize the pressurization chamber **6** to the desired trigger pressure. As soon as the trigger pressure is reached, the diaphragm **71** breaks, opening the passage opening **63**. The compressed gas in the pressurization chamber **6** is released and instantaneously occupies the receiving space **82** of the sabot **8** and applies a thrust force against its face, ejecting the sabot **8** and the drone **2** from the launch tube **3**, ejecting at the same time the tight plug **30**. Once ejected from the launch tube **3**, the sabot **8** separates into separate segments **80** as a result of air friction and the thrust exerted by the blades **22** of the drone **2** as they switch to the deployed position. Once at its maximum height, the drone **2** starts up and stabilizes, and its mission can begin.

(53) If it is desired to launch a plurality of drones **2**, a plurality of launch devices **1** according to the present invention can be mounted on the same vehicle (not shown). In this case, and if the pressure source is an external pressure source, a single pneumatic circuit is required to pressurize the pressurization chambers **6**. In this variant, each launch tube **3** can be colour-coded according to the payload **23** with which the drone **2** is equipped. This enables quick visual identification of the object to be ejected.

(54) It is understood that the particular embodiments just described are indicative and non-limiting, and that modifications may be made without departing from the scope of the present invention.

Claims

1. A device for pneumatic launching of a drone, the device comprising: a longitudinal launch tube having a first, rear end, which is closed, and a second, front end, which is open, and an interior of the launch tube defines a launch chamber intended to receive a drone and in which there is a seat against which the drone is positioned before the drone is launched; a pressurization chamber located at the rear end of the launch tube and capable of being pressurized by a pressure source, the pressurization chamber having a first, front end in which is provided a passage opening leading into the launch chamber and, in use, behind the drone placed against the seat; and temporary closing means for temporarily closing the passage opening, the temporary closing means being capable of switching instantaneously from a closed configuration, in which the passage opening is closed in a sealed manner, to an open configuration in which the passage opening is fully open; and the temporary closing means being arranged to switch from the closed configuration to the open configuration instantaneously under action of a gas pressure inside the pressurization chamber, after the pressurization chamber has reached a predetermined pressure, referred to as a trigger pressure, and a cross section of the passage opening being sufficiently large that, instantaneously with a switching of the temporary closing means from the closed configuration to the open configuration, the gas pressure prevailing in the pressurization chamber and a space located behind

the drone, then in communication with each other, is a desired launching pressure for the drone; and wherein the device comprises said pressure source, which is formed by an internal gas source arranged inside the pressurization chamber, said pressure source comprising, placed in the pressurization chamber, a diffuser and a second reactant, the diffuser delimiting a reservoir including a first reactant, the first and second reactants, when brought into contact with each other, reacting to generate a gas for pressurizing the pressurization chamber, a solenoid valve being arranged so as to allow selectively a fluid communication between the reservoir and the pressurization chamber.

2. The device according to claim 1, wherein the cross section of the passage opening is between 30% and 50% of a cross section of the launch chamber.

3. The device according to claim 1, wherein the temporary closing means are arranged to switch from the closed configuration to the open configuration automatically upon the gas pressure inside the pressurization chamber reaching the trigger pressure.

4. The device according to claim 3, wherein the temporary closing means comprises a rupture disc assembly that is disposed at the passage opening and comprises a calibrated diaphragm extending across the passage opening so as to close the passage opening, the calibrated diaphragm being dimensioned to break at the trigger pressure.

5. The device according to claim 4, wherein the rupture disc assembly is removably fixed to the pressurization chamber.

6. The device according to claim 5, wherein the rupture disc assembly comprises a support ring and a fixing ring of same dimensions and fixed to each other by fixing members passing through holes provided for this purpose in the support ring and the fixing ring, with a periphery of the calibrated diaphragm sandwiched between the support ring and the fixing ring, the rupture disc assembly being removably fixed to the front end of the pressurization chamber also by said fixing members.

7. The device according to claim 1, wherein the rear end of the launch tube is formed by a base removably fixed to an inner shoulder of the launch tube, from which base a tubular part extends, a longitudinal axis of the tubular part being coaxial with a longitudinal axis of the launch tube and the tubular part defining, together with the base, the pressurization chamber, a gasket being interposed between the tubular part and an inner annular edge of the inner shoulder.

8. The device according to claim 1, wherein the first reactant is hydrogen peroxide (H_2O_2) under pressure and the second reactant is a potassium permanganate (KMnO_4)-based solid catalyst.

9. The device according to claim 1, wherein the device further comprises a launch sabot capable of being received in the launch chamber and brought into contact against the seat, the launch sabot being configured to contain and surround the drone before launch and during the movement of the drone along the launch tube during ejection.

10. The device according to claim 9, wherein the launch sabot is formed of several separable segments.
